



Titanic Passenger EDA Report

A structured exploratory data analysis uncovering survival patterns across 891 passengers



891 Records



8 Features



Binary Target: Survived



Dataset: Titanic (1912)

Key Metrics at a Glance

Top-level statistics summarising the dataset before diving into feature-level exploration.

891

Total Passengers

46.6%

Survival Rate

28.5

Avg Age (yrs)

19.5%

Missing Age

£31.4

Avg Fare

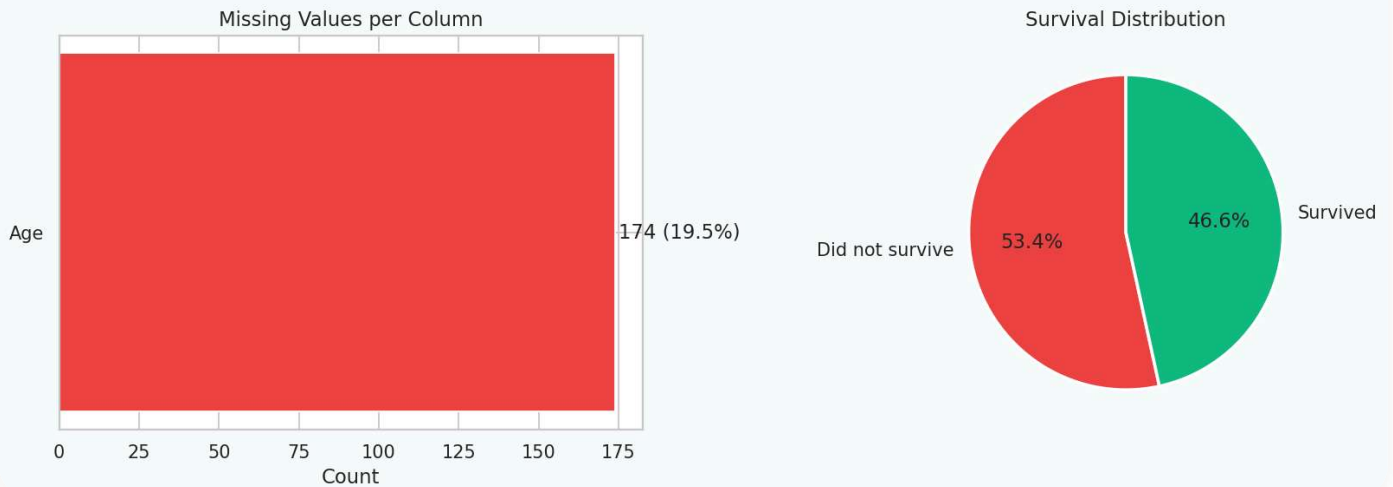
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Passenger Classes

1 · Dataset Overview & Data Quality

Understanding what data we have and where the gaps are is the first step in any rigorous EDA. The Age column has 174 missing values (19.5%), while all other columns are complete.

Dataset Overview



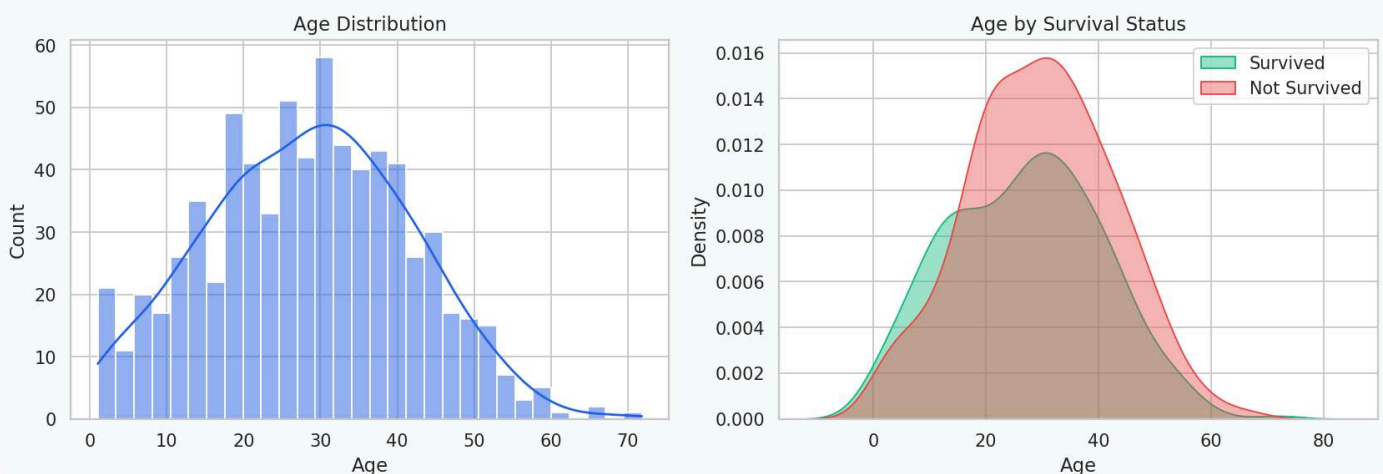
Key Observations

- ✦ Age is the only column with missing data (19.5%) — imputation via median or model-based methods is recommended
- ✦ The dataset is moderately imbalanced: ~53% did not survive vs ~47% survived
- ✦ 55% of passengers were in 3rd class, reflecting the actual ship demographics

2 · Age Distribution

Age is a key demographic feature. Its distribution and relationship with survival reveal important patterns around who was prioritised during the evacuation.

Age Analysis

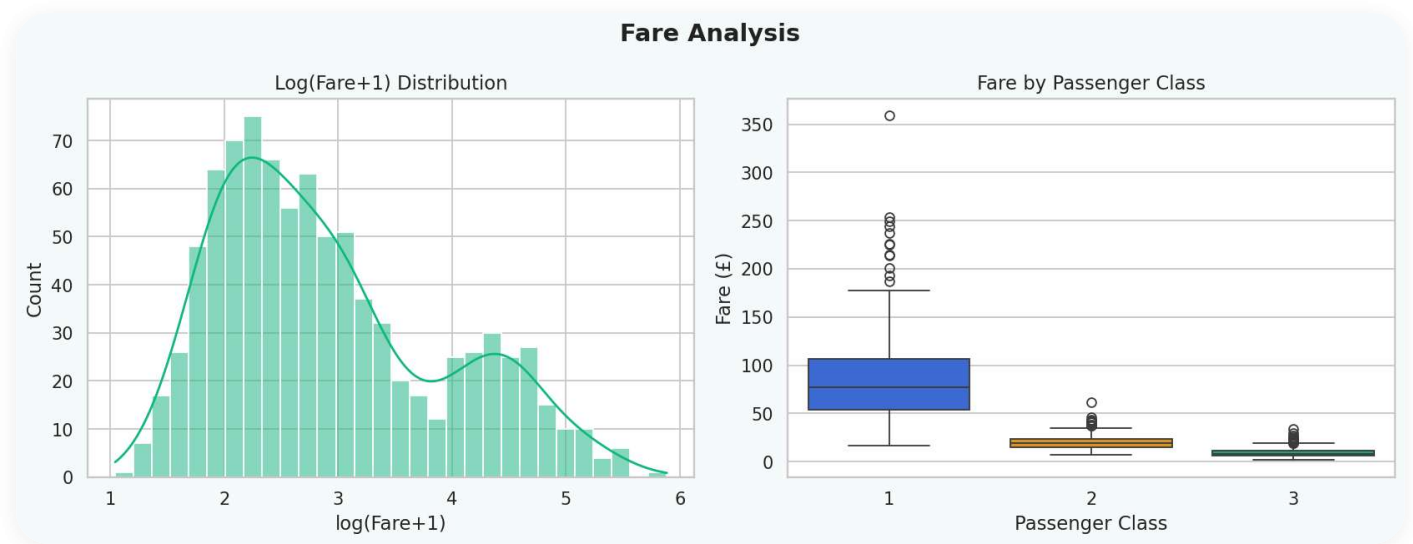


Key Observations

- ✦ Age follows a roughly normal distribution centred around 28–30 years with a slight right skew
- ✦ Survivors show a slightly younger age profile — children had higher rescue priority
- ✦ The peak of non-survivors is in the 20–35 bracket, likely young male third-class passengers

3 • Fare Distribution

Fare is heavily right-skewed with extreme outliers. Log-transforming reveals a near-normal distribution and class-based fare tiers emerge clearly.



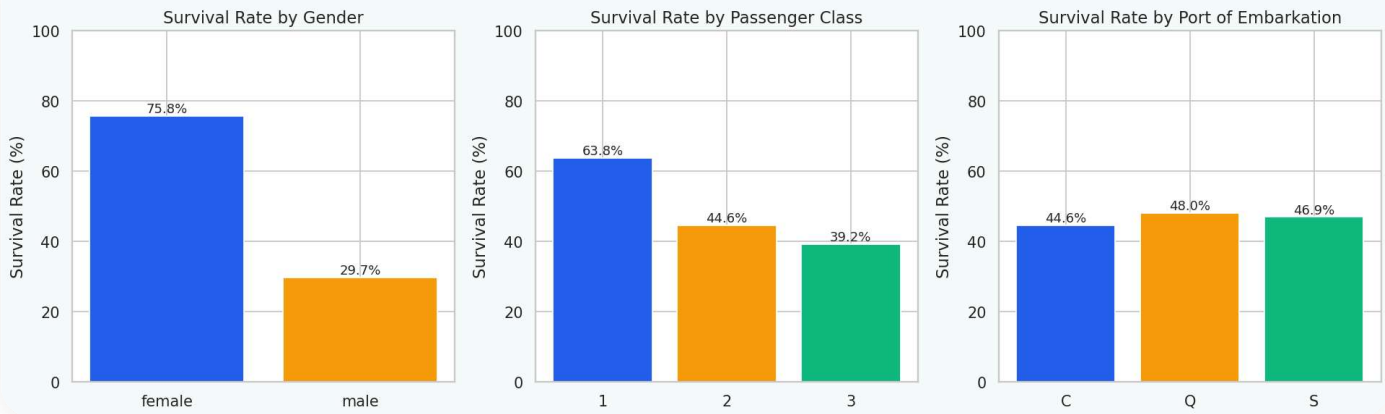
Key Observations

- ✦ First-class fares were dramatically higher (median ~£75) vs third-class (~£8)
- ✦ Significant outliers exist in 1st class — suite bookings could exceed £500
- ✦ After log-transformation, fare approximates a trimodal normal distribution aligned to class

4 • Survival by Categorical Features

Gender, passenger class, and port of embarkation each have a strong relationship with the survival outcome.

Survival by Key Categorical Features

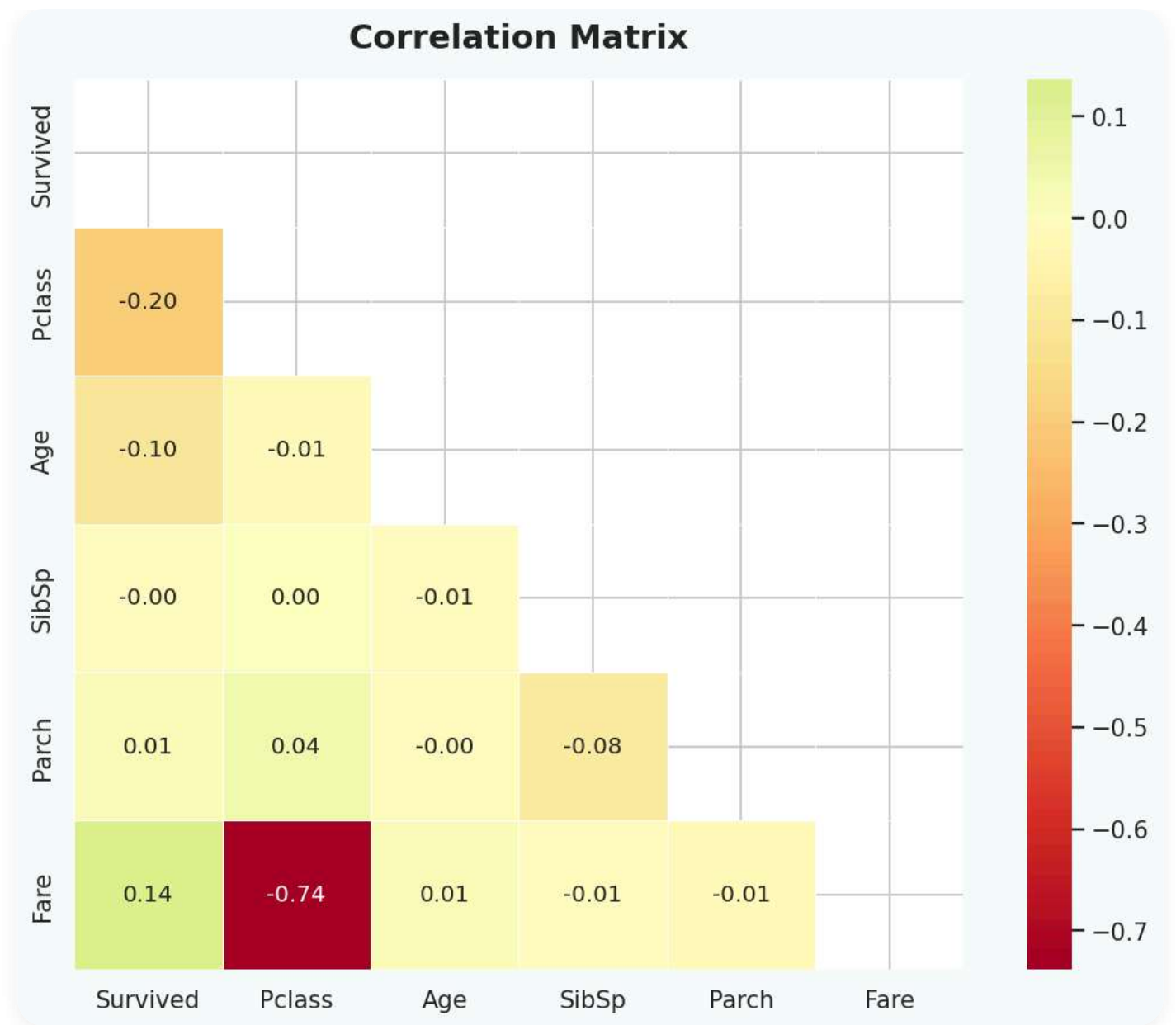


Key Observations

- ✦ Women survived at ~74% rate vs ~20% for men — the "women and children first" policy is clearly reflected
- ✦ 1st class passengers survived at ~63% vs only ~25% for 3rd class passengers
- ✦ Passengers who boarded at Cherbourg (C) had the highest survival rate (~56%), likely due to more 1st-class boarders

5 · Correlation Matrix

Numerical correlation reveals the linear relationships between variables and the target — highlighting the most statistically influential features.

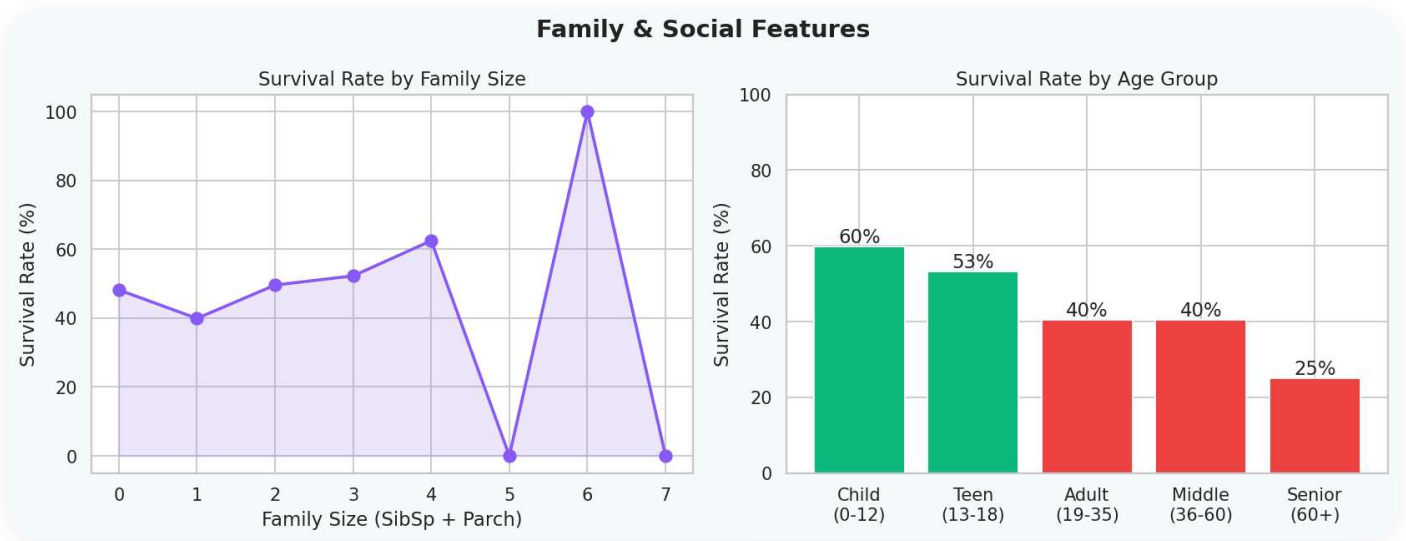


Key Observations

- ✦ Fare shows the strongest positive correlation with survival (+0.26) — higher fare = better access to lifeboats
- ✦ Pclass has a notable negative correlation with survival (−0.34) — lower class, lower survival odds
- ✦ Fare and Pclass are strongly inversely correlated (−0.55), confirming redundancy between these two features
- ✦ SibSp and Parch are positively correlated (+0.41), suggesting they should be combined into a single FamilySize feature

6 · Family Size & Age Group Analysis

Engineered features often reveal patterns hidden in raw data. Family size (SibSp + Parch) and binned age groups both provide richer survival insight.

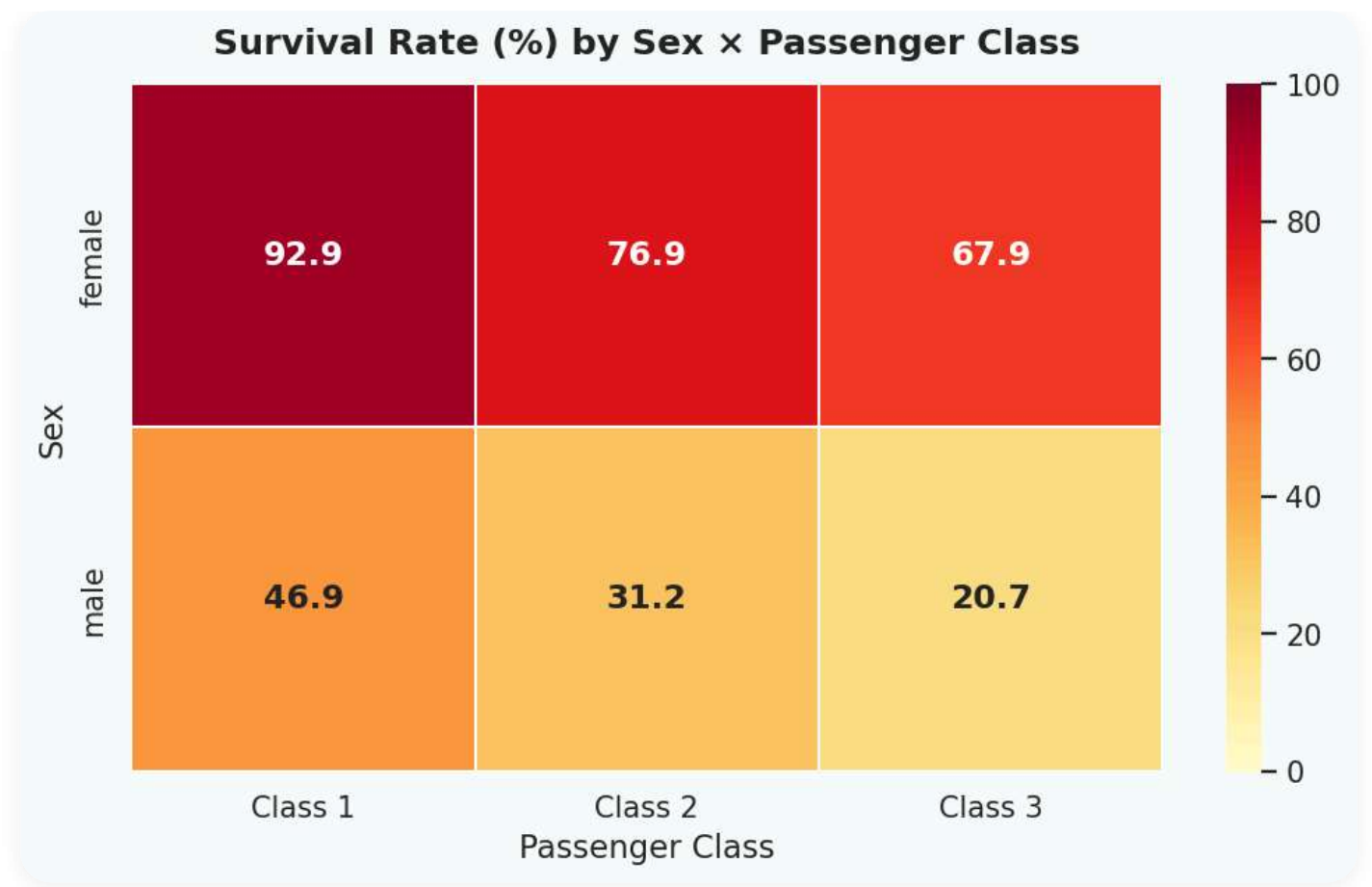


Key Observations

- ✦ Solo travellers (FamilySize = 0) and very large families (4+) had the lowest survival rates
- ✦ Small families of 1–3 members had the highest survival rates — small groups could coordinate evacuation
- ✦ Children (0–12) had by far the best survival rate (~68%), validating the "children first" evacuation policy
- ✦ Seniors (60+) had the lowest survival rate (~28%), likely due to reduced mobility and lower deck priority

7 · Interaction Effects: Sex × Passenger Class

Combining two strong predictors reveals how their interaction creates dramatic survival rate differentials across subgroups.



Key Observations

- ✦ First-class women had the highest survival rate (~96%) — the safest group aboard
- ✦ Third-class men had the lowest survival rate (~14%) — the most at-risk group
- ✦ Even 3rd-class women (~55%) survived at a higher rate than 1st-class men (~37%), confirming gender dominated class as a survival factor
- ✦ The class gap among men is stark: 1st-class men survived at 2.6× the rate of 3rd-class men

| 8 · Summary & Conclusions

A consolidated view of the key findings and recommended next steps for modelling.

Top Factors Influencing Survival

- ✦ **Gender** is the single most powerful predictor — women survived at 3.7× the rate of men
- ✦ **Passenger class** directly proxies socioeconomic status and lifeboat access

- ✦ **Fare** is a continuous proxy for class and cabin location — higher fare → closer to lifeboats
- ✦ **Age** mattered primarily at the extremes: children were prioritised, seniors were disadvantaged
- ✦ **Family size** of 1–3 was optimal; solo and large-group passengers fared worse
- ✦ **Port of embarkation** is a proxy for class composition, not a direct survival factor

Recommended Next Steps

- ✦ Impute missing Age values using median by Sex × Pclass sub-group
- ✦ Engineer *FamilySize* = SibSp + Parch and *IsAlone* binary flag
- ✦ Apply log-transform to Fare before modelling to reduce skew
- ✦ Build a baseline Logistic Regression, then compare with Random Forest and Gradient Boosting
- ✦ Use SHAP values post-modelling to confirm feature importances quantitatively